Rose, Blake (GE Research, US)

From: Dai, Jian (GE Research, US)
Sent: Friday, February 17, 2023 12:03 PM
To: Rose, Blake (GE Research, US)
Cc: Szczesny, Paul (GE Research, US)
Subject: Put application in to KR260 QSPI eMMC flash

Hi Blake,
This is how to boot and run your bare metal application.

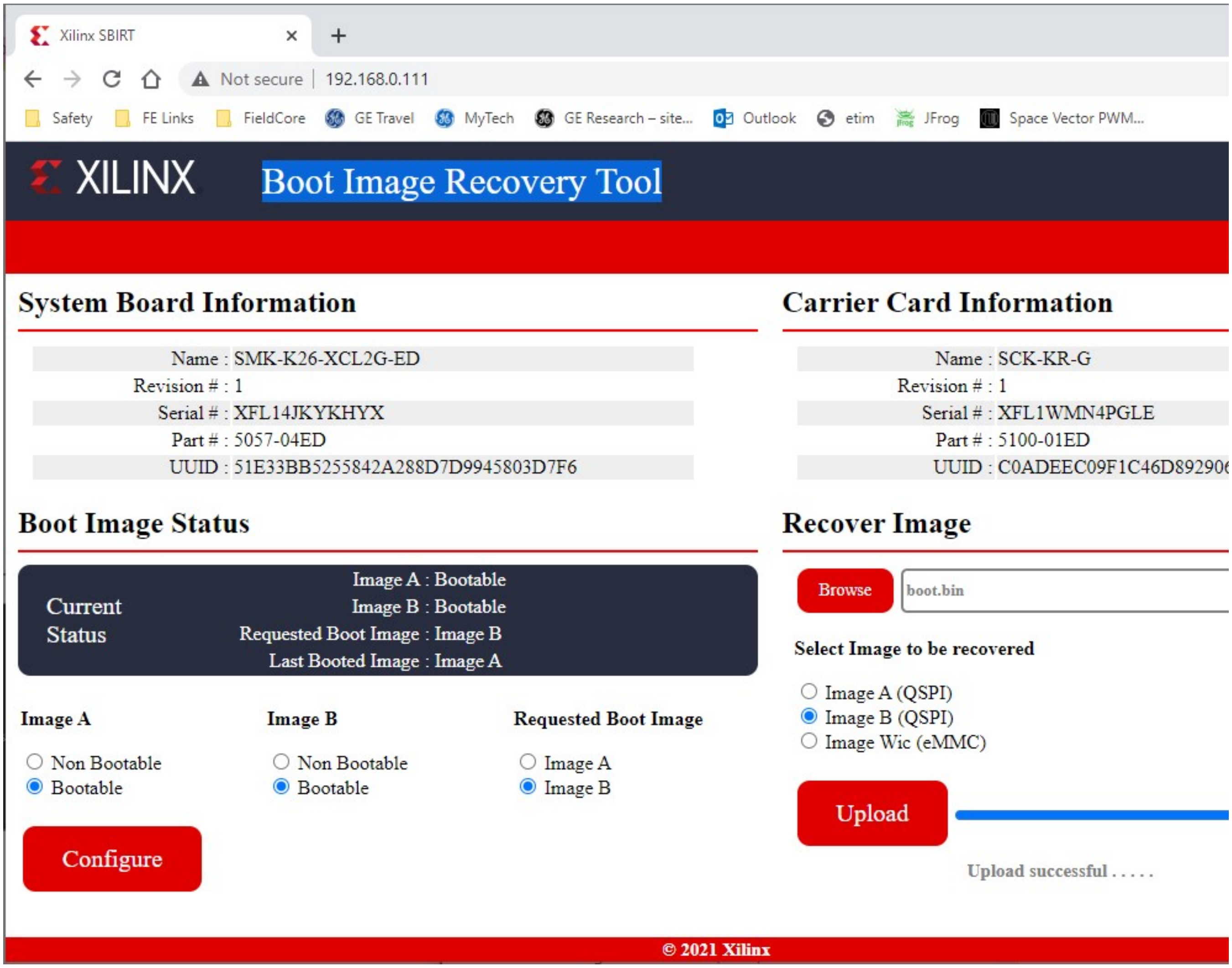
- Gather all the required files (pmufw.elf, fsbl.elf, < bit file exported by Vivado> and <you application elf file>) in one directory
- Create a bif file (kr260.bif, for ex.) in that directory with the following content:

```
the_ROM_image:
{
        [fsbl_config] a53_x64
        [pmufw_image]pmufw.elf
        [bootloader]fsbl.elf
        [destination_device=pl]kria_bd_wrapper.bit
        [destination_cpu=a53-0]a53_perf_test.elf

}
```

- Run the bootgen command:

```
Command Prompt

C:\projects\Vitis_Workspace\a53_perf_test\Debug>bootgen -arch zynqmp -image kr260.bif -o boot.bin


****** Xilinx Bootgen v2022.1
  **** Build date : Apr 18 2022-16:02:32
    ** Copyright 1986-2022 Xilinx, Inc. All Rights Reserved.

[WARNING]: [fsbl_config] a53_x64 | a53_x32 | r5_single | r5_dual is no more supported. Use 'destination_cpu

[INFO]   : Bootimage generated successfully


C:\projects\Vitis_Workspace\a53_perf_test\Debug>
```

- Use the KR260 built-in "Boot Image Recovery Tool" to update the flash
  Hold the "FWUEN" button while power on KR260. the serial port will show the IP address of the module. Use web browser to finish the update



- Reboot and your application is going to be running right away.

Jian